Package 'RJDProcessor'

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Description The rjdverse libraries are the officially recommended R software for seasonal adjustment in the European Central Bank and Statistical System. The RJDProcessor library integrates the rjdverse packages into a fully R-based production pipeline, ready to be used and easily extendable by methodologists. It offers the capability to manage the entire seasonal adjustment process: acquisition, processing, storage, automation, and not just seasonal adjustment of the data. Processing of multiple time series is possible by storing their specifications in JSON files, and interoperability with other JDemetra+ software is guaranteed because RJDProcessor can read workspaces and is able to produce them as an output.

RJDProcessor also provides functions to manage workspaces, such as splitting a workspace containing multiple time series into individual single-series workspaces, which are suitable for storing in databases with single time series records. Functions to merge workspaces are also available.

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Data_reader_ext_reg_tsplus.R
Data_reader_ext_reg_xlsx.R
Data_reader_ext_reg_csv.R
Data_reader_xlsx.R
Data_reader_list.R
Data_reader_xml.R

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Extended_tramoseats_spec.R
JD_JSON.R
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`check_data` *Compare the data in a workspace with only one time series with given data*

Description

This function compares the data in a workspace with only one time series with given raw data in array form

Usage

```
check_data(raw_data, ws_single_ts, raw_data_start = NA, raw_data_freq = NA)
```

Arguments

`raw_data`	Raw data to be compared with the ones in the workspace
`ws_single_ts`	The xml path of the workspace containing the data to be compared
`raw_data_start`	-optional- Default=NA, the starting date of the raw data in form "YYYY-MM-DD". If NA, the starting date is assumed to be the same as the workspace data
`raw_data_freq`	-optional- Default=NA, the frequency of the raw data (e.g. 12=monthly, 4=quarterly). If NA, the frequency is assumed to be the same as the workspace data

Value

Boolean: TRUE if data are the same, FALSE otherwise

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
single_workspaces_path <- "splitted_workspaces_to_merge"
single_ws<-load_workspace(paste0(single_workspaces_path,"\\VATPIC\\VATPIC.xml"))
result<-check_data(c(1,2,3), single_ws)
setwd(original_directory)
```

`check_external_regressors`
Verify the external regressors used in a workspace

Description

This function check whether the external regressors used in a workspace are up to date (i.e. they cover from the beginning to the end of the time series and have the same frequency). The workspace must contain a single time series.

Usage

```
check_external_regressors(ws_single_ts)
```

Arguments

ws_single_ts	The xml path of the workspace containing the data to be checked

Value

Boolean: TRUE if ckeck is ok, FALSE if there are problems in the external regressors.

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
single_workspaces_path <- "splitted_workspaces_to_merge"
single_ws<-load_workspace(paste0(single_workspaces_path,"\\VATPIC\\VATPIC.xml"))
compute(single_ws)
result<- check_external_regressors(single_ws)
setwd(original_directory)
```

---

compare_sa_ts *Compare the same models contained in two workspaces*

Description

This function compares the same models contained in two workspaces (old and new) plotting the respective sasonally adjusted series into a pdf file

Usage

```
compare_sa_ts(
  new_r_model = NA,
  new_model_workspace,
  old_model_workspace,
  materialized_ws_new = FALSE,
  materialized_ws_old = TRUE,
  java_processing_old_model = TRUE
)
```

Arguments

new_r_model	-optional- an R model obtained via RJDemetra::get_model(workspace). If != NA, this model is used as new model in the comparison. Default = NA, new_model_workspace is used as new model
new_model_workspace	the workspace (relative/absolute) path of the .xml file or object used as new model in the comparison

old_model_workspace
: the workspace (relative/absolute) path of the .xml file or object used as old model in the comparison

materialized_ws_new
: -optional- Default=FALSE, boolean field stating whether the new workspace in the comparison is passed as a matherialized or a virtual workspace (i.e. R object)

materialized_ws_old
: -optional- Default=TRUE, boolean field stating whether the new workspace in the comparison is passed as a matherialized or a virtual workspace (i.e. R object)

java_processing_old_model
: -optional- Default=TRUE, use Java models for the internal processing of the old workspace (faster than R)

Value

a "comparisons.pdf" file containig the plots of the seasonally adjusted time series of both the new and old workspaces. Series with the same series_name are reported in the same plot

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
input_workspace_materialized <- "workspace_test/workspace.xml"
virtual_workspace <- load_workspace(input_workspace_materialized)
compute(virtual_workspace)
compare_sa_ts(new_model_workspace = virtual_workspace,
              old_model_workspace = input_workspace_materialized ,
               materialized_ws_new=FALSE, materialized_ws_old=TRUE,
               java_processing_old_model=FALSE)
setwd(original_directory)
```

---

compare_workspaces *Compare two RJDemetra workspaces*

Description

This function compares two RJDemetra workspaces, either provided as objects or file paths, and generates a report of the differences in the final seasonally adjusted series and likelihood BIC values.

Usage

```
compare_workspaces(
  workspace1,
  workspace2,
  output_file = "differences_report.txt"
)
```

Arguments

workspace1	Either an RJDemetra workspace object or a file path to a workspace.
output_file	Character string specifying the path of the output report file. Default is "differences_report.txt".
patworkspace2	Either an RJDemetra workspace object or a file path to a workspace.

Details

The function loads the workspaces if they are provided as file paths, computes the models, and extracts the seasonally adjusted series and likelihood BIC. It then compares these values and records any differences in the output file.

Differences are checked for the last 36 observations of the seasonally adjusted series and the likelihood BIC values.

Value

The function does not return a value but writes a report to the specified file.

---

convert_numeric_matrix_to_mts

Convert a numeric matrix to an mts object

Description

This function takes a numeric matrix with dates as row names and series names as column names, and converts it to an mts (multivariate time series) object. It is goot for converting data read by data_readers and ext_reg_data_readers into mts that could be useful as input as usrdef.var for RJDemetra functions tramoseats() and X13()

Usage

```
convert_numeric_matrix_to_mts(data_matrix, freq = NULL)
```

Arguments

data_matrix	A numeric matrix where row names are dates in "YYYY-MM-DD" format, and column names are the names of the time series.
freq	-optional- The frequency of the time series. If not provided, the function will try to infer the frequency based on the difference between the first two dates.

Value

An mts object with the given time series data.

Examples

```
input_data_file_name <- system.file("extdata","CSV-FAS/grezzi_trim_FAS.csv", package = "RJDProcessor")
input_data_reader    <- Data_reader_csv(input_source = input_data_file_name)
data_matrix <- input_data_reader@read_data()

# Convert the matrix to an mts object
mts_object <- convert_numeric_matrix_to_mts(data_matrix)
print(mts_object)
```

`create_diagnostic_report1` *Create Diagnostic Report for Time Series Models*

Description

This function generates a diagnostic report for time series models, including BIC, p-values from Ljung-Box tests, and normality test results. It organizes the output into categories: all series, series with Ljung-Box problems, series with normality issues, and a summary of problematic series. Time series without a seasonal part are excluded from the diagnostics.

Usage

```
create_diagnostic_report1(
  workspace,
  output_file = "report.txt",
  exclude_non_seas = TRUE
)
```

Arguments

`workspace`	A workspace object containing the time series models.
`output_file`	A string specifying the output file path where the report will be saved (default is "report.txt").
`exclude_non_seas`	Logical. TRUE to exclude time series without a seasonal part from the diagnostics.

Examples

```
require(RJDemetra)

original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
ws_path <- "WorkspaceTUR-container/workspace-TUR.xml"

workspace <- load_workspace(file=ws_path)
create_diagnostic_report1(workspace, output_file = "report.out")
setwd(original_directory)
```

create_diagnostic_report2 *Create another Diagnostic Report for Time Series Models*

Description

This function generates a diagnostic report for time series models. It shows for every time series a complete view, including regression coefficients with their T-statistics as well as LB, LB2, Normality tests and BIC.

Usage

```
create_diagnostic_report2(
  workspace,
  output_file = "series_info.txt",
  exclude_non_seas = TRUE
)
```

Arguments

workspace	A workspace object containing the time series models.
output_file	A string specifying the output file path where the report will be saved (default is "report.txt").
exclude_non_seas	Logical. TRUE to exclude time series without a seasonal part from the diagnostics.

Examples

```
require(RJDemetra)

original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
ws_path <- "WorkspaceTUR-container/workspace-TUR.xml"

workspace <- load_workspace(file=ws_path)
create_diagnostic_report2(workspace, output_file = "report.out")
setwd(original_directory)
```

Data_reader_csv *Constructor (R-like) of the Data_reader object*

Description

This function creates a Data_reader object capable of reading data from CSV files and returning it using the `read_data()` function.

Usage

```
Data_reader_csv(input_source = NA, ...)
```

Arguments

input_source A string with file name (also with path).

Value

The Data_reader_csv object

Examples

```
input_data_file_name <- system.file("extdata","CSV-FAS/grezzi_trim_FAS.csv", package = "RJDProcessor")
input_data_reader    <- Data_reader_csv(input_source = input_data_file_name)
input_data_reader@read_data()
```

Data_reader_csv_istat_format

Constructor (R-like) of the Data_reader object

Description

This function creates a Data_reader object capable of reading data from CSV files in ISTAT format and returning it using the `read_data()` function. The ISTAT format is a csv file with dates in format YYYYqMM as rownames and time_series names as colnames

Usage

```
Data_reader_csv_istat_format(input_source = NA, ...)
```

Arguments

input_source A string with file name (also with path).

Value

The Data_reader_csv_istat_format object

Examples

```
input_data_file_name <- system.file("extdata","SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
input_data_reader <- Data_reader_csv_istat_format(input_source = input_data_file_name)
#input_data_reader@read_data()
```

Data_reader_ext_reg_csv

Constructor (R-like) of the Data_reader object

Description

This function creates a Data_reader_ext_reg object capable of reading data from CSV external regressors files and returning it using the `read_ext_reg_data()` function.

Usage

```
Data_reader_ext_reg_csv(input_source, ...)
```

Arguments

`input_source`	A string with the input: e.g. a file name (also with path) if the input is a file.

Value

The Data_reader_ext_reg_csv object

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_csv(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

Data_reader_ext_reg_tsplus

Constructor (R-like) of the Data_reader object

Description

This function creates a Data_reader_ext_reg object capable of reading data from TRAMO-SEATS+ external regressors files and returning it using the `read_ext_reg_data()` function.

Usage

```
Data_reader_ext_reg_tsplus(input_source, ...)
```

Arguments

`input_source`	A string with the input: e.g. a file name (also with path) if the input is a file.

Value

The Data_reader_ext_reg_tsplus object

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "SITIC-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_tsplus(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

Data_reader_ext_reg_xlsx

Constructor (R-like) of the Data_reader object

Description

This function creates a Data_reader_ext_reg object capable of reading data from XLSX external regressors files and returning it using the `read_ext_reg_data()` function.

Usage

```
Data_reader_ext_reg_xlsx(input_source, ...)
```

Arguments

`input_source`	A string with the input: e.g. a file name (also with path) if the input is a file.

Value

The Data_reader_ext_reg_tsplus object

Examples

```
require(RJDemetra)
input_workspace_xlsx    <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "SITIC-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xlsx)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_tsplus(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

Data_reader_list *Constructor (R-like) of the Data_reader object*

Description

This function creates a Data_reader object capable of reading data from a list and returning it using the `read_data()` function.

Usage

```
Data_reader_list(input_source = NA, ...)
```

Arguments

`input_source`	A string with file name (also with path).

Value

The Data_reader_csv object

Examples

```
FATEXP_10_list <- list("series_name"="FATEXP_10", "dates"=c("2005-01-01","2005-02-01","2005-03-01"),
                       "values"=c(12, 15, 11.1))
C_DEFL_list   <- list("series_name"="C_DEFL",   "dates"=c("2001-01-01","2001-02-01","2001-03-01"),
                       "values"=c(99, 100, 99.1))
# ...
input_data_list  <- list(FATEXP_10_list, C_DEFL_list) #add more time series if you want (here are 2)
input_data_reader <- Data_reader_list(input_source = input_data_list)
input_data_reader@read_data()
```

Data_reader_workspace *Constructor (R-like) of the Data_reader object*

Description

This function creates a Data_reader object capable of reading data from workspaces and returning them using the `read_data()` function.

Usage

```
Data_reader_workspace(input_source = NA, ...)
```

Arguments

`input_source`	A string with workspace xml file name (also with path).

Value

The Data_reader_workspace object

Examples

```
input_data_file_name <- system.file("extdata","WorkspaceTUR-container/workspace-TUR.xml", package = "RJDProc
input_data_reader    <- Data_reader_workspace(input_source = input_data_file_name)
mts <- input_data_reader@read_data()
```

Data_reader_xlsx *Constructor (R-like) of the Data_reader object*

Description

This function creates a Data_reader object capable of reading data from XLSX files and returning it using the `read_data()` function.

Usage

```
Data_reader_xlsx(input_source = NA, ...)
```

Arguments

`input_source`	A string with file name (also with path).

Value

The Data_reader_xlsx object

Examples

```
input_data_file_name <- system.file("extdata","XLSX-TUR/grezzi_trim_TUR.xlsx", package = "RJDProcessor")
input_data_reader    <- Data_reader_xlsx(input_source = input_data_file_name)
input_data_reader@read_data()
```

Data_reader_xml *Constructor (R-like) of the Data_reader object*

Description

This function creates a Data_reader object capable of reading data from XLSX files and returning it using the `read_data()` function.

Usage

```
Data_reader_xml(input_source = NA, ...)
```

Arguments

`input_source`	A string with file name (also with path).

Value

The Data_reader_xml object

Examples

```
input_data_file_name <- system.file("extdata","Prod.xml", package = "RJDProcessor")
# NOTE: absolute paths are better for this Data_reader
input_data_reader    <- Data_reader_xml(input_source = input_data_file_name)
#input_data_reader@read_data() # for reading the data
```

`from_full_to_reduced_JD_JSON_file` *Print a JSON file with only the fields that differ from the basic spec*

Description

This function prints a JSON string that contains only the fields of the JD_JSON object that differ from the ones of the basic specification (i.e. "RSA0, "RSA1", ...)

Usage

```
from_full_to_reduced_JD_JSON_file(
  JD_JSON_file,
  output_file_name = NA,
  indent = TRUE,
  basic_spec = NA
)
```

Arguments

`JD_JSON_file`	The name of the file in which the JD_JSON will be saved
`output_file_name`	-optional- The name of the file (optionally with path) in which the JD_JSON will be saved. If NA, the name of the file will be paste0(JD_DSON_file,"_reduced")
`indent`	-optional- Default TRUE. Print each field of the JSON in a different row
`basic_spec`	-optional- i.e. "RSA0", "RSA1", ... Default=NA. If NA, use the basic spec specified in the input JSON

Value

Void. A JSON file is saved on the filesystem

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory <- system.file("extdata", package = "RJDProcessor")
JSON_file_name <- system.file("extdata", "specifications_example3_full.txt",
                              package = "RJDProcessor")
setwd(extdata_directory)
from_full_to_reduced_JD_JSON_file(JSON_file_name, "specification_new_out2.txt")
setwd(original_directory)
```

from_reduced_to_full_JD_JSON_file

Print a JSON file with all the JD_JSON fields explicit

Description

This function prints a JSON string that contains all the fields of the JD_JSON object explicitly defined.

Usage

```
from_reduced_to_full_JD_JSON_file(
  JD_JSON_file,
  output_file_name = NA,
  indent = TRUE
)
```

Arguments

JD_JSON_file	The name of the file in which the JD_JSON will be saved
output_file_name	-optional- The name of the file (optionally with path) in which the JD_JSON will be saved. If NA, the name of the file will be paste0(JD_DSON_file,"_full")
indent	-optional- Default TRUE. Print each field of the JSON in a different row

Value

Void. A JSON file is saved on the filesystem with all the JD_JSON fields

Examples

```
original_directory <- getwd()
extdata_directory <- system.file("extdata", package = "RJDProcessor")
JSON_file_name <- system.file("extdata", "specifications_example2.txt", package = "RJDProcessor")
setwd(extdata_directory)
from_reduced_to_full_JD_JSON_file(JSON_file_name)
setwd(original_directory)
```

get_r_model_from_j_model

Get an R list with model information from java model

Description

This function gets an R list with model information from java model

Usage

```
get_r_model_from_j_model(j_model)
```

Arguments

j_model	a Java model obtained from RJDemetra

Value

a list containing all the models specifications

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
input_workspace <- "workspace_test/workspace.xml"
virtual_workspace <- load_workspace(input_workspace)
compute(virtual_workspace)
m <- get_jmodel(virtual_workspace)
r_model_list <- get_r_model_from_j_model(m)
setwd(original_directory)
```

---

get_single_ts_workspaces

Get a single workspace for each series in a workspace

Description

This function gets a single time series workspace for each time series SA model present in a given workspace

Usage

```
get_single_ts_workspaces(
  full_workspace,
  single_workspaces_path,
  compressed_ws = TRUE,
  clean_single_ws_directory = TRUE,
  from_TS_PLUS_plugin = TRUE
)
```

Arguments

full_workspace	A workspace with multiple time series SA models
single_workspaces_path	The path in which the single workspcaes will be stored
compressed_ws	-optional- Default=TRUE, compress the single workspaces that will be created
clean_single_ws_directory	-optional- Default=TRUE, delete the original workspace after creating the single series ones
from_TS_PLUS_plugin	-optional- Default=TRUE, the original workspace is created by the SA_ext_plugin. In that case some internal adjustments are needed and they could make the computation slower. There are no problems if this field is TRUE but the ws is created without SA_ext_plugin

Value

A set of materialized workspaces in single_workspaces_path, one workspace for each time series SA model contained in full_workspace

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
single_workspaces_path <- "splitted_workspaces"
full_workspace         <- "WorkspaceTUR-container\\workspace-TUR.xml"
compressed_ws          <-  FALSE
get_single_ts_workspaces(full_workspace, single_workspaces_path,
                         compressed_ws = compressed_ws )
setwd(original_directory)
```

JD_JSON_file_processor

Process a JD_JSON file

Description

This function processes a JSON file with JD_JSON fields and returns a virtual workspace

Usage

```
JD_JSON_file_processor(
  input_data_reader,
  ext_reg_data_reader,
  spec_file_name,
  output_workspace_dir = NA,
  series_to_proc_names = NA,
  java_processing = TRUE
)
```

Arguments

input_data_reader	A specific Data_reader object (CSV, XLSX, ...) to read the input
ext_reg_data_reader	A specific Data_reader_ext_reg object (CSV, XLSX, ...) to read the external regrssors
spec_file_name	Name of the file (with path, if desired) containing the JD_JSON to be processed (e.g. "specification_new.txt")
output_workspace_dir	-optional- Name of the directory that will contain the output workspace. Default=NA stores the workspace in a directory called "output_workspace_container"
series_to_proc_names	-optional- vector of names of time series to be processed (e.g. c('VATASA','VATPIA')) . Default=NA: all the series are processed

java_processing
: -optional- Default=TRUE. Use only Java API (faster) for the internal computation

Value

a virtual workspace, already processed

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
spec_file_name <- "specifications_to_proc.txt"
input_workspace_directory <- "WorkspaceTUR-container/workspace-TUR.xml"
input_data_file_name      <- "CSV-TUR/grezzi_trim_TUR.csv"
regr_directory            <- "CSV-TUR/regr"
diff <- TRUE # Reduced JSON if diff=TRUE, Full JSON format otherwise

############################## Operational flow ##############################

input_data_reader <- Data_reader_csv(input_source = input_data_file_name)
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)
JD_JSON_from_materialized_workspace(input_workspace_directory,
         ext_reg_input_data_reader, JSON_file_name = spec_file_name,
         diff=TRUE, java_processing=FALSE)
series_to_proc_names <- NA #c("FATEXP_13", "C_DEFL") # NA to process all
virtual_workspace <- JD_JSON_file_processor(input_data_reader = input_data_reader,
         ext_reg_data_reader = ext_reg_input_data_reader,
         spec_file_name = spec_file_name,
         output_workspace_dir = "output_workspace_container",
         series_to_proc_names = series_to_proc_names,
         java_processing = FALSE)
# set java_processor=TRUE to speed-up the operations, but it does not work with
# workspaces readed by sa-ext plugin
m <- get_model(virtual_workspace) #get directly the R model (slower)
from_reduced_to_full_JD_JSON_file(spec_file_name)
compare_sa_ts(new_model_workspace = virtual_workspace,
              old_model_workspace = input_workspace_directory ,
              materialized_ws_new=FALSE, materialized_ws_old=TRUE,
              java_processing_old_model=FALSE)
setwd(original_directory)
```

JD_JSON_from_materialized_workspace

Turn model spec of a materialized workspace in JD_JSON

Description

This function represent model specifications contained into a materialized workspace in JD_JSON

Usage

```
JD_JSON_from_materialized_workspace(
  workspace,
  ext_reg_input_data_reader,
  regr_directory = NA,
  JSON_file_name = "JD_JSON_specification.txt",
  diff = TRUE,
  java_processing = FALSE
)
```

Arguments

workspace	Name of the workspace xml file (also with path).
ext_reg_input_data_reader	A Data_reader_ext_reg object, to read the external regressors in the desired format (csv, xlsx, tramoseats+, ...)
regr_directory	-optional- Name of the directory containing the sources (e.g. files) of the external regressors
JSON_file_name	-optional- Name of the JSON file to be created. If NA the file will be called "JD_JSON_specification.txt"
diff	-optional- if TRUE a reduced version of the JSON specification is produced; In the reduced version, fields with default values equals to the ones of the default specification (i.e. "RSA0", "RSA1", ...) are not reported Default=TRUE
java_processing	-optional- If TRUE, the function works internally with Java API (faster), otherwise it uses R API. Default=FALSE
input_data_reader	A Data_Reader object

Value

A JSON file saved on the filesystem

Examples

```
require(RJDemetra)

input_workspace <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                          package = "RJDProcessor")
input_data_file_name      <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv",
                                          package = "RJDProcessor")
regr_directory        <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
diff <- TRUE   # Reduced JSON if diff=TRUE, Full JSON format otherwise

input_data_reader         <- Data_reader_csv(input_source = input_data_file_name)
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)
JD_JSON_from_materialized_workspace(input_workspace, ext_reg_input_data_reader, JSON_file_name = "specificat
```

`JD_JSON_from_virtual_workspace` *Turn model spec of a virtual (R) workspace in JD_JSON*

Description

This function represent model specifications contained into an R workspace in JD_JSON

Usage

```
JD_JSON_from_virtual_workspace(
  ws,
  ext_reg_input_data_reader,
  JSON_file_name = "JD_JSON_specification.txt",
  diff = TRUE,
  java_processing = TRUE
)
```

Arguments

`ws`	workspace R object.
`ext_reg_input_data_reader`	A Data_reader_ext_reg object, to read the external regressors in the desired format (csv, xlsx, tramoseats+, ...)
`JSON_file_name`	-optional- Name of the JSON file to be created. If NA the file will be called "JD_JSON_specification.txt"
`diff`	-optional- if TRUE a reduced version of the JSON specification is produced; In the reduced version, fields with default values equals to the ones of the default specification (i.e. "RSA0", "RSA1", ...) are not reported Default=TRUE
`java_processing`	-optional- If TRUE, the function works internally with Java API (faster), otherwise it uses R API. Default=TRUE
`input_data_reader`	A Data_Reader object
`regr_directory`	-optional- Name of the directory containing the sources (e.g. files) of the external regressors

Value

A JSON file saved on the filesystem

Examples

```
require(RJDemetra)

input_workspace_directory <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                         package = "RJDProcessor")
input_data_file_name      <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv",
                                         package = "RJDProcessor")
regr_directory        <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
```

```
diff <- TRUE   # Reduced JSON if diff=TRUE, Full JSON format otherwise

input_data_reader         <- Data_reader_csv(input_source = input_data_file_name)
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)

original_directory <- getwd()
extdata_directory <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)

ws <- load_workspace(file = input_workspace_directory)
JD_JSON_from_virtual_workspace(ws, ext_reg_input_data_reader, JSON_file_name = "specifications_new_out.txt",
setwd(original_directory)
```

`JD_JSON_to_materialized_workspace` *Turn a JD_JSON in a materialized workspace*

Description

This function obtain a JD_JSON file from a workspace stored in the filesystem (in a directory). See test foder for examples

Usage

```
JD_JSON_to_materialized_workspace(
  workspace_dir = NA,
  JSON_file,
  input_data_reader,
  ext_reg_data_reader = NA,
  series_to_proc_names = NA
)
```

Arguments

`workspace_dir`	-optional- the directory of the input workspace. Default = NA stores the workspace in a directory called "output_workspace_container"
`JSON_file`	Name of the JSON file that will be produced (also with path).
`input_data_reader`	A Data_Reader object
`ext_reg_data_reader`	-optional- A Data_reader_ext_reg object, to produce the metadata of the external regressors (i.e. create the names of the csv rather than a xlsx files o other containers for external regressors) Default = NA, do not consider external regressors (discouraged)
`series_to_proc_names`	-optional- an array containing the name of the series to be included in the workspace among the ones present in the JD_JSON file e.g. c("VATASA","VATAIA")

Value

void in R environment, a workspace materialized in the filesystem

Examples

```
require(RJDemetra)
input_JD_JSON     <- system.file("extdata", "specifications_example1.txt", package = "RJDProcessor")
input_data_file_name <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv", package = "RJDProcessor")
regr_directory       <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
input_data_reader         <- Data_reader_csv(input_source = input_data_file_name)
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)
JD_JSON_to_materialized_workspace(workspace_dir="ws_out" ,input_JD_JSON, input_data_reader, ext_reg_input_da
```

`JD_JSON_to_virtual_workspace`

Turn a JD_JSON in a virtual workspace

Description

This function obtain a virtual workspace from a JD_JSON file. See test foder for examples

Usage

```
JD_JSON_to_virtual_workspace(
  JSON_file,
  input_data_reader,
  ext_reg_data_reader = NA,
  series_to_proc_names = NA
)
```

Arguments

`JSON_file`	Name of the JSON file to turn in a workspace (also with path).
`input_data_reader`	A Data_Reader object
`ext_reg_data_reader`	-optional- A Data_reader_ext_reg object, to read the external regressors in the desired format (csv, xlsx, tramoseats+, ...) Default = NA, do not consider external regressors (discouraged)
`series_to_proc_names`	-optional- an array containing the name of the series to be included in the workspace among the ones present in the JD_JSON file e.g. c("VATASA","VATAIA")

Value

A virtual workspace

Examples

```
require(RJDemetra)
input_JD_JSON     <- system.file("extdata", "specifications_example1.txt", package = "RJDProcessor")
input_data_file_name <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv", package = "RJDProcessor")
regr_directory       <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
input_data_reader         <- Data_reader_csv(input_source = input_data_file_name)
```

```
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)
ws <- JD_JSON_to_virtual_workspace(input_JD_JSON, input_data_reader, ext_reg_input_data_reader, series_to_pr
```

`merge_workspaces` *Merge many workspaces in one*

Description

This function gets merges many workspaces contained in a given folder into one workspace

Usage

```
merge_workspaces(
  source_workspaces_path,
  merged_ws_name = "merged_ws",
  merged_ws_dir = NA,
  compressed = TRUE,
  delete_originals = TRUE,
  silent = TRUE
)
```

Arguments

`source_workspaces_path`	The path in which to find all the workspcaes that will be merged
`merged_ws_name`	-optional- Default="merged_ws". The name of the workspace that will be created
`merged_ws_dir`	-optional- Default=NA, the path in which the merged workspace will be stored. If NA it will be stored in the current directory
`compressed`	-optional- Default=TRUE, workspaces to be merged are compressed
`delete_originals`	-optional- Default=TRUE, delete original workspaces after merging them
`silent`	-optional- Default=TRUE, do not print status bar and messages during the operations

Value

A virtual (R) and a marerialized workspace containing all the multiprocessings and time series SA models of the original workspaces

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
single_workspaces_path <- "splitted_workspaces_to_merge"
full_workspace_path    <- "merged_ws_test"
compressed_ws          <-  FALSE
ws_merged <- merge_workspaces(source_workspaces_path=single_workspaces_path,
```

```
              merged_ws_dir = full_workspace_path, compressed = TRUE,
              delete_originals = FALSE, silent=TRUE)

setwd(original_directory)
```

---

produce_fully_R_workspace *Process a JD_JSON file*

Description

This function processes a JSON file with JD_JSON fields and returns a virtual workspace

Usage

```
produce_fully_R_workspace(
  ext_reg_input_data_reader = NA,
  original_ws_xml,
  new_r_ws_folder
)
```

Arguments

ext_reg_input_data_reader	A specific Data_reader_ext_reg object (CSV, XLSX, ...) to read the external regressors. Default=NA.
original_ws_xml	xml file (also with path) of the workspace to be transformed into a fully R workspace
new_r_ws_folder	Name of the directory that will contain the output, fully R, workspace.

Value

a virtual workspace, already processed

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
original_ws_xml <- "WorkspaceTUR-container/workspace-TUR.xml"
new_r_ws_folder <- "output_fully_R_ws"
regr_directory  <- "SITIC-TUR/regr"
ext_reg_input_data_reader <- Data_reader_ext_reg_tsplus(regr_directory)
produce_fully_R_workspace(ext_reg_input_data_reader, original_ws_xml, new_r_ws_folder)


extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
original_ws_xml <- "WorkspaceFAS-standard-container//FAS.xml"
new_r_ws_folder <- "output_fully_R_ws"
```

```
regr_directory  <- "CSV-FAS//regr"
ext_reg_input_data_reader <- Data_reader_ext_reg_csv(regr_directory)
produce_fully_R_workspace(ext_reg_input_data_reader, original_ws_xml, new_r_ws_folder)
setwd(original_directory)
```

---

read_data,Data_reader_csv-method

Get the data from a Data_reader_csv

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_csv'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYY-MM-DD) and colnames = time series names (string)

Examples

```
input_data_file_name <- system.file("extdata","CSV-FAS/grezzi_trim_FAS.csv", package = "RJDProcessor")
input_data_reader    <- Data_reader_csv(input_source = input_data_file_name)
input_data_reader@read_data()
```

---

read_data,Data_reader_csv_istat_format-method

Get the data from a Data_reader_csv_istat_format

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_csv_istat_format'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYYqMM) and colnames = time series names (string)

Examples

```
input_data_file_name <- system.file("extdata","SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
input_data_reader    <- Data_reader_csv_istat_format(input_source = input_data_file_name)
input_data_reader@read_data()
```

---

read_data,Data_reader_list-method
Get the data from a Data_reader_list

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_list'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYY-MM-DD) and colnames = time series names (string)

Examples

```
FATEXP_10_list <- list("series_name"="FATEXP_10", "dates"=c("2005-01-01","2005-02-01","2005-03-01"),
                       "values"=c(12, 15, 11.1))
C_DEFL_list   <- list("series_name"="C_DEFL",   "dates"=c("2001-01-01","2001-02-01","2001-03-01"),
                       "values"=c(99, 100, 99.1))
#...
input_data_list  <- list(FATEXP_10_list, C_DEFL_list) #add more time series if you want (here are 2)
input_data_reader <- Data_reader_list(input_source = input_data_list)
input_data_reader@read_data()
```

---

read_data,Data_reader_workspace-method
Get the data from a Data_reader_workspace

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_workspace'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYY-MM-DD) and colnames = time series names (string)

Examples

```
input_data_file_name <- system.file("extdata","WorkspaceTUR-container/workspace-TUR.xml", package = "RJDProc
input_data_reader    <- Data_reader_workspace(input_source = input_data_file_name)
mts <- input_data_reader@read_data()
```

read_data,Data_reader_xlsx-method

Get the data from a Data_reader_csv

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_xlsx'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYY-MM-DD) and colnames = time series names (string)

Examples

```
input_data_file_name <- system.file("extdata","XLSX-TUR/grezzi_trim_TUR.xlsx", package = "RJDProcessor")
input_data_reader    <- Data_reader_xlsx(input_source = input_data_file_name)
input_data_reader@read_data()
```

read_data,Data_reader_xml-method

Get the data from a Data_reader_xml

Description

This function returns the data from the input_source of the object.

Usage

```
## S4 method for signature 'Data_reader_xml'
read_data(object, ...)
```

Value

data in form of numeric matrix, with rownames = dates (in string format, YYYY-MM-DD) and colnames = time series names (string)

Examples

```
input_data_file_name <- system.file("extdata","Prod.xml", package = "RJDProcessor")
# NOTE: absolute paths are better for this Data_reader
input_data_reader    <- Data_reader_xml(input_source = input_data_file_name)
#input_data_reader@read_data() # for reading the data
```

`read_ext_reg_data,Data_reader_ext_reg_csv-method`
Read external regressors data

Description

This function reads data from external regressors and returns it as a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Usage

```
## S4 method for signature 'Data_reader_ext_reg_csv'
read_ext_reg_data(
  object,
  var_info = NULL,
  time_series_info = NULL,
  frequency = NA_integer_,
  ...
)
```

Arguments

`var_info`	A string with file name (also with path).
`time_series_info`	A string with time series name in workspace name (also with path).
`frequency`	i.e. 12 = monthly data, 4 = quarterly data

Value

a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_csv(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

read_ext_reg_data,Data_reader_ext_reg_tsplus-method
Read external regressors data

Description

This function reads data from external regressors and returns it as a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Usage

```
## S4 method for signature 'Data_reader_ext_reg_tsplus'
read_ext_reg_data(
  object,
  var_info = NULL,
  time_series_info = NULL,
  frequency = NA_integer_,
  ...
)
```

Arguments

var_info	A string with file name (also with path).
time_series_info	A string with time series name in workspace name (also with path).
frequency	i.e. 12 = monthly data, 4 = quarterly data

Value

a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "SITIC-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_tsplus(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

`read_ext_reg_data,Data_reader_ext_reg_xlsx-method`
Read external regressors data

Description

This function reads data from external regressors and returns it as a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Usage

```
## S4 method for signature 'Data_reader_ext_reg_xlsx'
read_ext_reg_data(
  object,
  var_info = NULL,
  time_series_info = NULL,
  frequency = NA_integer_,
  ...
)
```

Arguments

`var_info`	A string with file name (also with path).
`time_series_info`	A string with time series name in workspace name (also with path).
`frequency`	i.e. 12 = monthly data, 4 = quarterly data

Value

a numeric matrix with variable names as colnames and YYYY-MM-DD dates as rownames

Examples

```
require(RJDemetra)
input_workspace_xlsx    <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "XLSX-TUR/grezzi_trim_TUR.xlsx", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "XLSX-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xlsx)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_xlsx(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
vars_matrix         <- data_reader_ext_reg@read_ext_reg_data(all_model_ext_vars_info, "VATASC",
                                                             frequency=12)
```

read_ext_reg_info,Data_reader_ext_reg_csv-method

Read information about external regressors from a workspace

Description

This function returns a list of information about external regressors used in the models contained in a workspaces

Usage

```
## S4 method for signature 'Data_reader_ext_reg_csv'
read_ext_reg_info(object, var_info_container, adjust_path = TRUE, ...)
```

Arguments

var_info_container
: workspace xml file path

Value

list() of information about external regressors

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "CSV-TUR/grezzi_trim_TUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "CSV-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_csv(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
```

read_ext_reg_info,Data_reader_ext_reg_tsplus-method

Read information about external regressors from a workspace

Description

This function returns a list of information about external regressors used in the models contained in a workspaces

Usage

```
## S4 method for signature 'Data_reader_ext_reg_tsplus'
read_ext_reg_info(object, var_info_container, adjust_path = TRUE, ...)
```

Arguments

var_info_container
: workspace xml file path

Value

list() of information about external regressors

Examples

```
require(RJDemetra)
input_workspace_xml     <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "SITIC-TUR/grezziTUR.csv", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "SITIC-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xml)
compute(ws)
data_reader_ext_reg       <- Data_reader_ext_reg_tsplus(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
```

---

read_ext_reg_info,Data_reader_ext_reg_xlsx-method

Read information about external regressors from a workspace

Description

This function returns a list of information about external regressors used in the models contained in a workspaces

Usage

```
## S4 method for signature 'Data_reader_ext_reg_xlsx'
read_ext_reg_info(object, var_info_container, adjust_path = TRUE, ...)
```

Arguments

var_info_container
: workspace xml file path

Value

list() of information about external regressors

Examples

```
require(RJDemetra)
input_workspace_xlsx    <- system.file("extdata", "WorkspaceTUR-container/workspace-TUR.xml",
                                       package = "RJDProcessor")
input_data_file_name    <- system.file("extdata", "XLSX-TUR/grezzi_trim_TUR.xlsx", package = "RJDProcessor")
regr_directory        <- system.file("extdata", "XLSX-TUR/regr", package = "RJDProcessor")
ws                        <- load_workspace(file = input_workspace_xlsx)
compute(ws)
```

```
data_reader_ext_reg       <- Data_reader_ext_reg_xlsx(regr_directory)
all_model_ext_vars_info <- data_reader_ext_reg@read_ext_reg_info(ws)
```

`seasonality_changes_report` *Report dei cambiamenti nella stagionalità tra due workspace*

Description

Confronta i modelli ARIMA di due workspace (vecchio e nuovo) e identifica le serie che hanno cambiato il loro stato di stagionalità (passando da modelli stagionali a non stagionali e viceversa).

Usage

```
seasonality_changes_report(
  workspace_old,
  workspace_new,
  output_file = "seasonality_changes.txt"
)
```

Arguments

`workspace_old`	Oggetto workspace di RJDemetra precedente.
`workspace_new`	Oggetto workspace di RJDemetra attuale.
`output_file`	Percorso del file .txt dove verrà salvato il report.

Value

La funzione non restituisce oggetti in R, ma scrive un file di testo.

`update_data` *Update the data of a workspace*

Description

This function update the data of a workspace's time series basing on the data read by a Data_reader object already initialized. The time series read by the Data_reader must have the same colnames as the time series names of the workspace to produce an update

Usage

```
update_data(workspace_xml_path, data_reader)
```

Arguments

`workspace_xml_path`	Path of the xml file of the workspace whose data have to be updated
`data_reader`	A Data_reader object already initialized with its input source

Examples

```
require(RJDemetra)
original_directory <- getwd()
extdata_directory  <- system.file("extdata", package = "RJDProcessor")
setwd(extdata_directory)
ws_xml_path  <- "TUR_ws_test_container/merged_ws.xml"
dr <- RJDProcessor::Data_reader_csv(input_source = "rawdata_TUR.csv")
# num_mat<-dr@read_data() # to check if the data are available
update_data(ws_xml_path, dr)
setwd(original_directory)
```

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